

Total No. of Questions : 4]

SEAT No. :

PD24

[Total No. of Pages : 2

[6409]-221

S.E. (Computer Engg./AI & DS/Computer Science) (Insem)

DATA STRUCTURES AND ALGORITHMS

(2019 Pattern) (Semester - IV) (210252)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Construct a hash table to insert following data in hash table. Use quadratic probing for collision resolution. Hash function $key \% 10$. Consider one slot per bucket. **[6]**

Elements: 13, 48, 25, 33, 55, 72, 147, 43

Calculate the average number of comparisons.

b) Explain the folding method of hashing with examples. **[5]**

c) Explain the terms Key density and load density **[4]**

Calculate the load density for the hash table with 20 slots, that stores 1000 elements

OR

Q2) a) Consider the hash table with 9 buckets. The hash function is $h(K) = k \bmod \text{table size}$. The collision is resolved by chaining using replacement. The following keys are inserted in the order: 5, 37, 19, 24, 29, 23, 48, and 17. Calculate the maximum chain length. **[6]**

b) Explain the concept of skip list with example. State its application. **[5]**

c) Compare static hashing and extendible hashing. **[4]**

P.T.O.

- Q3)** a) Generate Huffman tree and find the codes for following alphabets having specified frequency: A-4, B-5, C-2, D-7, E-3, F-1 [6]
b) Write an algorithm for non recursive post order traversal of binary tree. [5]
c) Explain how to convert general tree to binary tree with example. [4]

OR

- Q4)** a) Construct a binary tree from its given preorder traversal PQSXYTRUV and postorder traversal XYSTQUVRP and apply inorder traversal to it. [6]
b) Write an algorithm to delete a node from binary search tree. [5]
c) Explain threaded binary tree with example and state its advantages. [4]

